

ANALYTIX PLATFORM

Enterprise Informatics & Discovery Platform - Comprehensive User Manual

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1. Introduction & Overview

The AnalytiX Platform is a state-of-the-art enterprise informatics and in silico drug discovery suite. It integrates advanced cheminformatics, bioinformatics, federated data queries, and AI-driven scientific assistance into a single unified platform. This document serves as the official operational manual and compliance reference for AnalytiX. It details all 20 active platform modules, illustrating their purpose, business value, scientific value, user interface layout, step-by-step workflows, expected results, common errors, troubleshooting, and best practices. Designed for both research scientists and system administrators, this manual provides the instructions necessary to operate the platform in a validated FDA 21 CFR Part 11 compliant environment.

1.1 Document Control & Version History

The system maintenance ledger records modifications to this manual:

Version	Date	Author	Changes Summary
1.0.0	Jan 2025	S. Kurapati	Initial system documentation template.
2.0.0	Jun 2025	S. Kurapati	Added cheminformatics and bioinformatics modules.
3.0.0	Dec 2025	S. Kurapati	Added compliance, E-Signatures, and data lineage.
4.0.0	Jun 2026	Compliance Div	FDA 21 CFR Part 11 alignment and AI Scientist prompts.

2. Platform Architecture Overview

AnalytiX platform implements a modern, secure, and distributed architecture. It isolates application domains across nine distinct PostgreSQL database schemas managed by independent microservices. Below is a summary of the schemas and ports:

Service Module	Port	Schema Name	Data Objects / Primary Tables
Authentication Service	8001	gen_auth	users, roles, permissions, audit_auth
Metadata Catalog Service	8002	metadata	entities, values, attribute_fields (EAV)
Query Builder Service	8003	query	query_history, query_templates
Cheminformatics Service	8004	cheminformatics	compounds, structures, chemical_scaffolds
Connector Service	8005	connector	connectors, schemas, schedules
Audit & Compliance Service	8006	audit	audit_logs, electronic_signatures
Data Lineage Service	8007	lineage	lineage_nodes, lineage_edges
Bioinformatics Service	8008	bio	sequences, alignments, clusters
Workflow Automation Service	8009	workflow	workflow_definitions, workflow_runs
AI Scientist Copilot Service	8010	ai	chat_sessions, chat_messages, SQL_grounding

3. User Roles & Permissions

Access to the AnalytiX platform is governed by Role-Based Access Control (RBAC) to ensure security and compliance with FDA 21 CFR Part 11 guidelines. The system defines three primary default roles: 1. **Administrator (Admin)**: Full control over system configurations, user administration, connector management, RBAC settings, and database maintenance. 2. **Scientist**: Full access to research modules, including Data Registry, Compound Explorer, Bioinformatics Hub, Query Builder, Analytics Workbench, Workflow Designer, and AI Scientist Copilot. 3. **Compliance Officer**: Special access to the Audit Trail, Compliance Console, electronic signature verification, and system validation reports.

4. Platform Modules Detail

This section provides a detailed operational breakdown of all 20 modules.

Module 1: Dashboard & Informatics Hub

Purpose: Provides the primary administrative and scientific launchpad of the AnalytiX platform. Renders real-time platform key performance indicators (KPIs), active computational pipeline statuses, recent project activity, and immediate database connector health feeds.

Business Value: Minimizes decision lag by providing operations and scientific managers a unified overview of all active drug discovery workloads, compute resource availability, and system throughput metrics.

Scientific Value: Integrates heterogeneous study telemetry (e.g., active target structure counts, compound registration velocities, alignment queue sizes) to help researchers immediately identify platform usage and pipeline blockages.

Navigation Instructions: Navigate to the dashboard by selecting the 'Dashboard' link from the sidebar menu (or accessing the '/' or '/dashboard' path).

Screen Overview: Divided into three zones: a top-row KPI panel (4 metric cards), a middle section containing a split grid for 'Compute Pipeline Status' and 'System Feeds', and a right-hand quick shortcut sidebar.

Component Explanation:

Component / UI Element	Description & Functional Actions
Platform Health Card	A visual status indicator reporting backend connection status, server response times, and active database connections.
Active Pipelines Panel	A live table reporting active screening runs, workflow executions, and bioinformatics alignment queues with status indicators.
Active User Activity Feed	A real-time list of system audit actions completed by researchers (e.g., compound registrations, queries run).
Quick Navigation Cards	A set of tiles linking to Compound Explorer, Visual Query Builder, and Bioinformatics Hub.

Step-by-Step Workflow Steps:

1. Log into the AnalytiX platform using your verified electronic credentials.
2. Verify the 'Secure Session' indicator in the top header is green and active.
3. Review the 'Platform Health Card' for active connector database configurations.
4. Assess the running counts of compound libraries, active workflows, and sequence files in the KPI ribbon.
5. Double-click on any active pipeline status card to drill down into the respective module detail page.

Expected Results: Immediate rendering of active metrics, live connection status of backend databases, and high-fidelity navigation links.

Errors & Troubleshooting:

Common Error Scenario	Troubleshooting Solution
Database Connectivity Latency	If a service is restarting, a yellow 'LATENCY' badge appears. Check connection parameters in the Settings tab.
API Handshake Failed	Red status badge. Verify VPN connections or consult your system administrator.

Tips & Best Practices: Utilize the Dashboard as an operational health check. Keep the browser window open to receive real-time, event-driven updates from the message broker.

Module 2: Data Registry

Purpose: Serves as the unified registration repository for all newly discovered or purchased chemical compounds, biological sequences, and assay structures. Enforces strict schema validations and uniqueness constraints before inserting records.

Business Value: Protects IP integrity and ensures a single version of truth for molecular and biological inventory, preventing costly assay duplicates.

Scientific Value: Ensures that SMILES and FASTA inputs are validated via cheminformatics/bioinformatics libraries (RDKit, BioPython) to prevent corrupted molecular structures from polluting database records.

Navigation Instructions: Click the 'Data Registry' pill button in the top header navigation panel or access the '/metadata' catalog and click 'Register Entity'.

Screen Overview: A multi-tab workspace with tabs for: 'Chemical Registration' (SMILES/MOL inputs), 'Sequence Registration' (FASTA formats), and 'Assay Data Loader' (CSV/Excel parsing).

Component Explanation:

Component / UI Element	Description & Functional Actions
SMILES Text Field	Input area for SMILES strings with real-time chemical valence and sanity checks.
Structure Canvas Previewer	Renders molecular structural formulas on the fly using vector graphics (SVGs) dynamically compiled by RDKit.
FASTA File Drag-Drop Area	Accepts uploads of raw FASTA files containing biological sequences with automatic sequence type detection (DNA/RNA/Protein).
Register Button	Triggers synchronous backend validation, audits, database insertion, and electronic signature prompts.

Step-by-Step Workflow Steps:

1. Navigate to the Data Registry via the header button.
2. Select the 'Chemical Registration' tab.
3. Paste the SMILES string 'CC(=O)NC1=CC=C(O)C=C1' (Acetaminophen) into the SMILES input field.
4. Confirm the vector drawing matches the expected structural formula.
5. Enter molecular metadata fields (e.g., Lot Number, Purity, Storage Condition).
6. Click 'Register' and authenticate with your digital signature credentials.

Expected Results: A notification popup indicating 'Entity Registered successfully' with a unique internal identifier (e.g., GQC-100204) and corresponding audit log entry.

Errors & Troubleshooting:

Common Error Scenario	Troubleshooting Solution
Invalid SMILES Representation	RDKit parse failure. Verify chemical structure and format syntax.
Duplicate Compound Detected	An identical chemical structure is already present under another ID. The system highlights the existing ID link.

Tips & Best Practices: For high-throughput screening libraries, use the batch upload template (CSV format) under the 'Assay Data Loader' tab to register up to 10,000 compounds at once.

Module 3: Metadata Catalog

Purpose: A flexible, dynamic repository implemented via an Entity-Attribute-Value (EAV) database model that stores all properties, descriptors, annotations, and parameters for compounds, sequences, and assays without requiring hardcoded database migrations.

Business Value: Enables rapid extension of the database schema as new discovery methodologies and properties emerge, reducing IT overhead.

Scientific Value: Allows researchers to tag compounds with arbitrary custom attributes (e.g., custom IC50 limits, ADMET prediction metrics, gene expression scores) for subsequent visual filtering.

Navigation Instructions: Select 'Metadata Catalog' from the left-hand navigation sidebar (or navigate directly to '/metadata').

Screen Overview: Features an left-hand hierarchical taxonomy browser and a right-hand details table representing all active metadata definitions, datatypes, and validation constraints.

Component Explanation:

Component / UI Element	Description & Functional Actions
Taxonomy Tree View	Displays folders and tags representing metadata classes (e.g., ADMET, Physicochemical, Assay Results).
Attribute Table List	Shows defined attributes (e.g., mw, clogp, ic50_nm, smiles) with datatype badges (Float, String, Structure).
New Attribute Dialog	A modal window allowing administrators to add new metadata fields with custom bounds and validation regex.
Search Metadata Box	Filters attributes by name, category, or description.

Step-by-Step Workflow Steps:

1. Access the Metadata Catalog from the sidebar.
2. Search for the attribute 'clogp' using the Search box.
3. Click on the 'clogp' row in the table to display validation limits, target ranges, and description fields.
4. Click 'Edit Attribute' (requires Compliance Officer or Administrator role).
5. Modify the minimum value constraint to -2.0 and maximum value to 9.0.
6. Click 'Save Attributes' and review the change confirmation details.

Expected Results: Immediate propagation of the modified metadata bounds, updated database validation schema, and audit history entry.

Errors & Troubleshooting:

Common Error Scenario	Troubleshooting Solution
Schema Validation Mismatch	Occurs when trying to change a data type of an attribute that already has active values. Delete or migrate values
Permission Denied	Read-only access. Request Metadata administrator access permissions.

Tips & Best Practices: Always define clear physical units (e.g., nM, g/mol, angstroms) in the description of metadata attributes to ensure clarity during query operations.

Module 4: Connector Management

Purpose: Manages physical connections, credentials, schema definitions, and synchronization tasks for external databases (Snowflake, PostgreSQL, Oracle, SQL Server, MongoDB) and scientific systems (Benchling, LabWare ELN, LIMS).

Business Value: Breaks down internal data silos by federating access to third-party databases, eliminating manual data export/import workflows.

Scientific Value: Enables live integration of raw screening data from active robotic liquid-handlers and sequencing machines straight into the research workbench.

Navigation Instructions: Click 'Data Connectors' or 'Enterprise Integrations' in the left-hand sidebar menu, or navigate to '/connectors'.

Screen Overview: A dashboard displaying cards for each configured connector, connection status pills (Connected, Syncing, Offline), and buttons for initiating syncs or editing credentials.

Component Explanation:

Component / UI Element	Description & Functional Actions
Connector Inventory Grid	Cards containing connector name, type (e.g., Snowflake), last sync timestamp, and status badges.
New Connector Button	Launches a step-by-step wizard for database/API settings (host, port, DB, username, credentials).
Schema Discovery View	Renders the external database schema, tables, columns, and keys mapped during metadata discovery.
Sync Scheduler panel	Configures cron jobs for automated metadata sync runs.

Step-by-Step Workflow Steps:

1. Open the Connectors page from the sidebar.
2. Click 'Add New Connector' (launches wizard).
3. Choose 'PostgreSQL' from the connector types list.
4. Enter connection parameters: Host 'localhost', Port '5432', Database 'assay_db', Username 'app_read'.
5. Toggle 'Encrypted Credentials' to enabled, and enter Password.
6. Click 'Test Connection' to verify network and credentials.
7. Click 'Discover Schema' to map external tables.

8. Click 'Save Connector'.

Expected Results: A new connector card is added to the dashboard, displaying a green 'CONNECTED' status, and schema details become queryable in the metadata catalog.

Errors & Troubleshooting:

Common Error Scenario	Troubleshooting Solution
Connection Timeout	Verify database port accessibility through local firewall rules and VPN tunnels.
Authentication Failure	Invalid password or database privileges. Request read access from database administrator.

Tips & Best Practices: Use dedicated, read-only database accounts for all connectors to minimize security risks and audit footprints.

Module 5: Compound Explorer

Purpose: A powerful chemical search and visualization dashboard designed to execute exact structures, substructures, and molecular similarity searches against millions of registered compounds using RDKit query engines.

Business Value: Shortens compound screening loops, helping chemists identify relevant chemical space and structural analogs within minutes.

Scientific Value: Allows researchers to execute structural searches with Tanimoto coefficient metrics and view structural overlays to identify pharmacophores.

Navigation Instructions: Click 'Compound Explorer' in the left sidebar or go to '/compounds'.

Screen Overview: Features an interactive JSME chemical structure drawer panel on the left, query configurations in the center, and a results grid showing structure cards on the right.

Component Explanation:

Component / UI Element	Description & Functional Actions
JSME Chemical Structure Drawer	Web-based chemical sketcher to draw molecules, export SMILES, or paste MOL files.
Search Mode Dropdown	Selects between 'Exact Structure', 'Substructure', or 'Similarity (Tanimoto)'.
Similarity Threshold Slider	Controls the Tanimoto similarity threshold percentage (range: 50% to 100%).
Results Structure Grid	Renders structural formulas of matching compounds with key descriptors (MW, LogP, PSA, HBD, HBA) and check

Step-by-Step Workflow Steps:

1. Open Compound Explorer from the sidebar.
2. In the JSME Structure Drawer, sketch a benzene ring and add an amide substituent.
3. Select 'Substructure Search' from the dropdown.
4. Click the 'Execute Search' button.
5. Wait for results to render. Use the checkboxes to select molecules of interest.
6. Click the 'Export to CSV' button to save results locally.

Expected Results: Renders a paginated grid of all compounds containing the sketched amide-benzene pharmacophore, along with live calculated physicochemical values.

Errors & Troubleshooting:

Common Error Scenario	Troubleshooting Solution
Empty Canvas Error	No structure drawn. Draw a chemical scaffold or paste a SMILES string before executing structure search.
RDKit Initialization Failure	WebAssembly file could not load. Reload the browser tab to restart the cheminformatics backend.

Tips & Best Practices: When executing similarity searches, a threshold of 80% is the industry standard for finding meaningful lead candidates while filtering out noise.

Module 6: Bioinformatics Explorer

Purpose: A comprehensive sequence analytics center designed to parse, align, cluster, and visualize biological sequences (DNA, RNA, proteins) using BioPython adapters and visual matrices.

Business Value: Reduces dependency on disconnected command-line bioinformatics utilities, facilitating sequence-structure correlations in a single platform.

Scientific Value: Allows biologists to identify motifs, run global/local sequence alignments (Needleman-Wunsch / Smith-Waterman), and build sequence similarity dendrograms.

Navigation Instructions: Click 'Bioinformatics Hub' in the left-hand navigation sidebar (or go to '/bioinformatics').

Screen Overview: A dashboard with cards linking to the 'Sequence Database', 'Sequence Alignment Studio', 'Sequence Clustering Center', and 'Sequence Explorer'.

Component Explanation:

Component / UI Element	Description & Functional Actions
Sequence Import Field	Input text box for pasting sequences in FASTA/Raw formats, or uploading sequence files (.fasta, .gb).
Alignment Workspace Panel	Displays color-coded aligned sequences with consensus markers and gap statistics.
Clustering Tree / Dendrogram	Visualizes hierarchical clustering results using interactive Plotly tree maps and clusters.
Motif Search Input	Searches sequences for specific conservation patterns (e.g., DNA primer sequences or protein binding pockets).

Step-by-Step Workflow Steps:

1. Open the Bioinformatics Hub and click 'Sequence Alignment Studio'.
2. Paste two FASTA protein sequences in the input field.
3. Select 'Global (Needleman-Wunsch)' alignment algorithm.
4. Set gap opening penalty to -10 and gap extension to -1.
5. Click 'Perform Alignment'.
6. View the alignment output matrix and review the sequence identity score.

Expected Results: A side-by-side aligned sequence grid highlighting matching bases (green), substitutions (yellow), and insertions/deletions (red dashes).

Errors & Troubleshooting:

Common Error Scenario	Troubleshooting Solution
Unsupported Sequence Type	Mixed sequence types (e.g. aligning protein with DNA). Ensure both sequences are of the same molecule type.
Fasta Format Error	Missing header starting with '>' character. Wrap the sequence with correct FASTA metadata.

Tips & Best Practices: Use local alignment for longer genomic sequences to identify small regions of high conservation without penalizing long mismatching ends.

Module 7: Query Builder

Purpose: Provides a visual query interface to query complex, federated databases across chemical structures, biological targets, metadata attributes, and assay metrics without requiring SQL knowledge.

Business Value: Empowers scientists to perform self-service data mining, reducing reliance on database administrators and data science queues.

Scientific Value: Combines structural constraints (e.g., substructure matching) with numerical descriptors (e.g., molecular weight) and assay targets (e.g., target proteins) in a single query matrix.

Navigation Instructions: Click 'Query Builder' or 'Visual Query Builder' in the sidebar or go to '/query-builder'.

Screen Overview: A node-based query workspace with block categories (Filters, Entities, joins) on the left, a central canvas to arrange query blocks, and a query SQL/JSON review panel on the bottom.

Component Explanation:

Component / UI Element	Description & Functional Actions
Query Block Canvas	A visual space where users drag and drop filter blocks (e.g., 'Molecular Weight < 500', 'IC50 < 10nM') and connect them.
Add Rule Dropdown	Selects attributes from the Metadata Catalog to instantiate a new query block.
Query History Panel	Accesses and loads queries run previously by the current researcher.
Execute Query Button	Compiles the visual AST node graph to SQL, runs it against the federated engine, and redirects to the results table.

Step-by-Step Workflow Steps:

1. Navigate to the Visual Query Builder.
2. Drag a new 'Rule Block' into the canvas.
3. Select 'molecular_weight' from the attribute dropdown, choose operator '<', and input value '500'.
4. Click 'Add Group' to introduce an AND condition block.
5. Add another rule with 'clogp' operator '<' and value '5.0'.
6. Click 'Run Query' in the bottom toolbar.

Expected Results: Compilation of the query graph, database execution, and rendering of a table displaying matching compounds.

Errors & Troubleshooting:

Common Error Scenario	Troubleshooting Solution
Orphaned Node Error	A logical block is unconnected on the canvas. Connect all query blocks with valid relational wires.
Query Timeout Error	The query was too complex or returned too many rows. Refine rules to restrict chemical or metadata ranges.

Tips & Best Practices: Save complex queries as Templates using the 'Save Template' button to share query configurations with other team members.

Module 8: Analytics Workbench

Purpose: A comprehensive scientific data science sandbox equipped to perform 4-Parameter Logistic (4PL) regression for dose-response curves, Principal Component Analysis (PCA), t-SNE clustering, K-Means clustering, and Pearson correlation matrices.

Business Value: Reduces licensing fees for third-party analysis packages by hosting advanced statistics directly on the server next to the data registry.

Scientific Value: Enables researchers to identify structural clusters in high-dimensional screening data and generate publication-ready plots directly from queries.

Navigation Instructions: Select 'Analytics Workbench' from the sidebar menu, or navigate to '/analytics-workbench'.

Screen Overview: A split layout featuring statistical parameter settings on the left (e.g., dimensions, clusters, fit algorithms) and interactive, high-fidelity Plotly charts on the right.

Component Explanation:

Component / UI Element	Description & Functional Actions
Analysis Type Selector	Dropdown to choose between 4PL Regression, PCA, t-SNE, K-Means, or Correlation Matrix.
Input Data Selector	Allows selection of query results or registered assay tables as input datasets.
Interactive Plot Canvas	Renders statistical charts (dose-response curves, scatter plots, Heatmaps) using Plotly with tooltips, zooming, and pan.
Export Plot Button	Downloads high-resolution vector figures in PDF or PNG formats.

Step-by-Step Workflow Steps:

1. Open the Analytics Workbench.
2. Select the 'Dose-Response (4PL)' tab.
3. Choose 'Assay Data Series 12' as the input dataset.
4. Map 'Concentration (uM)' to the X-axis and 'Inhibition (%)' to the Y-axis.
5. Click 'Fit 4PL Model'.
6. Review the calculated EC50/IC50 value and hill slope coefficient.

Expected Results: A fitted Sigmoidal curve plotted over the concentration data points, with statistical metrics (R2, IC50) displayed in the summary card.

Errors & Troubleshooting:

Common Error Scenario	Troubleshooting Solution
Insufficient Data Points	Fewer than 4 concentration points provided. 4PL models require at least 4 unique concentrations to fit.
Model Convergence Failure	Data is too noisy or lacks a sigmoidal shape. Check concentrations or select a linear fit.

Tips & Best Practices: Use PCA or t-SNE before clustering to reduce chemical descriptor dimensionality, allowing K-Means to identify more cohesive pharmacophore clusters.

Module 9: Workflow Automation Engine

Purpose: A node-based drag-and-drop designer and engine that automates scientific pipelines, orchestrating data retrieval, ADMET prediction, molecular docking, and electronic sign-off into standard operating procedures.

Business Value: Ensures process standardization and eliminates manual tasks, accelerating compound design cycles.
Scientific Value: Automates multi-stage workflows such as: taking query output -> running ADMET predictions -> filtering by Lipinski rules -> running molecular docking -> saving results to the database.

Navigation Instructions: Click 'Workflow Designer' in the left-hand sidebar menu, or navigate to '/workflows'.

Screen Overview: A large grid canvas where node blocks representing tasks (Database Query, RDKit, Docking, Approval, E-Signature) are connected by directional wires representing execution flow.

Component Explanation:

Component / UI Element	Description & Functional Actions
Workflow Canvas	The visual design area where execution pipelines are constructed and reviewed.
Node Toolbox	A sidebar menu listing available scientific, logical, and database nodes.
Node Configurator Drawer	Slide-out panel where properties of the selected node are configured.
Execute Pipeline Button	Saves and initiates execution of the workflow definition, displaying live progress.

Step-by-Step Workflow Steps:

1. Open Workflow Designer.
2. Drag a 'Database Query' node and a 'Lipinski Filter' node onto the canvas.
3. Connect the output of the Query node to the input of the Filter node.
4. Drag a 'Molecular Docking' node and connect the Filter output to it.
5. Click the 'Database Query' node, open the configurator, and select the query template 'EGFR Candidates'.
6. Click 'Run Workflow' and track the green execution path.

Expected Results: A step-by-step progress tracking indicator, concluding with a run execution report showing results and a download link for docked structures.

Errors & Troubleshooting:

Common Error Scenario	Troubleshooting Solution
Relational Type Mismatch	The output type of node A is incompatible with the input of node B. Verify data formats in the configurator.
Execution Timeout	Molecular docking took longer than the timeout threshold. Adjust docking parameters to reduce grid size.

Tips & Best Practices: Insert 'Approval / E-Signature' nodes before write operations to enforce regulatory compliance in automated workflows.

Module 10: Audit Trail & Compliance

Purpose: Records all user actions, database mutations, configuration modifications, and signature approvals inside a secure ledger. Each log entry is cryptographically linked to the previous entry using SHA-256 hash chains, creating a tamper-evident audit history.

Business Value: Ensures legal defense and audit readiness for FDA reviews, protecting clinical submissions from disqualification.

Scientific Value: Enforces complete reproducibility of in silico experiments by tracing exactly who performed a computation, what parameters were used, and when it occurred.

Navigation Instructions: Click 'Audit Trail' in the sidebar or go to '/admin/audit'.

Screen Overview: Shows a table of historical system logs with filters for date range, user, action type, and status, and a prominent 'Verify Ledger Integrity' button.

Component Explanation:

Component / UI Element	Description & Functional Actions
Audit Logs Table	Lists timestamp, user email, action, module, IP address, and cryptographic block hash.
Ledger Integrity Panel	Displays the result of the SHA-256 chain verification, reporting 'VALID' or warning of unauthorized modification.
Action Type Filter	Dropdown to isolate specific auditable events (e.g., LOGIN, DATA_INSERT, DELETE, E_SIGN).
Export Audit Log Button	Generates a signed CSV or PDF report of the current log selection.

Step-by-Step Workflow Steps:

- 1. Navigate to the Audit Trail page.
- 2. Filter actions by setting the 'Module' to 'Cheminformatics'.
- 3. Select a specific log entry detailing a compound modification.
- 4. Click the log entry to open the JSON detail payload showing 'Before' and 'After' states.
- 5. Click 'Verify Ledger Integrity' to run the verification algorithm across the database table.

Expected Results: Verification report showing 'All audit ledger blocks successfully validated. Hash chain is unbroken.' with green checkmarks.

Errors & Troubleshooting:

Common Error Scenario	Troubleshooting Solution
Chain Integrity Warning	Red alert showing a hash mismatch. Indicates a database record was altered outside the platform interface. Cont
Signature Verification Failure	Public keys do not match. Verify the user's compliance certificate.

Tips & Best Practices: Schedule weekly ledger integrity checks to automatically run and email reports to the Compliance Officer.

Module 11: FDA 21 CFR Part 11 Compliance

Purpose: Enforces specific regulations for electronic records and signatures, including session timeouts, credential double-entry, password expiration policies, and restricted admin actions.

Business Value: Guarantees regulatory compliance for electronic records, enabling digital workflows to replace physical signatures.

Scientific Value: Validates the digital authenticity of calculations and target selections, ensuring results can be included in FDA Investigational New Drug (IND) applications.

Navigation Instructions: Access the 'Compliance Console' from the left sidebar and select the 'FDA Part 11' tab.

Screen Overview: A dashboard displaying compliance checklists, policy configuration controls, and validation settings.

Component Explanation:

Component / UI Element	Description & Functional Actions
Session Timeout Field	Configures inactivity duration (in minutes) before automatic session termination.
Double-Entry Toggle	Enforces re-entering password for critical actions (e.g., compound approval, data export).
Compliance Status Panel	Real-time list of CFR Part 11 requirements showing green (Met) or red (Action required) markers.
Validation Report Builder	Generates system validation logs for auditor inspection.

Step-by-Step Workflow Steps:

1. Open the Compliance Console and go to the 'FDA Part 11' tab.
2. Enable 'Double-Factor E-Signatures' for all metadata schema changes.
3. Set the 'Inactivity Session Timeout' to 15 minutes.
4. Click 'Apply Compliance Settings'.
5. Test the setting by attempting to export a query dataset; verify that a password confirmation dialog appears.

Expected Results: Security configuration update, audited system event, and enforcement of security constraints across all user sessions.

Errors & Troubleshooting:

Common Error Scenario	Troubleshooting Solution
Session Terminated	Session expired due to inactivity. Re-authenticate to resume work.
Verification Lockout	Too many incorrect password entries during electronic signing. Account is locked for 30 minutes.

Tips & Best Practices: Conduct monthly simulated compliance audits to verify that user permissions and activity logs meet FDA requirements.

Module 12: Electronic Signatures

Purpose: A secure digital signing engine that prompts users to authenticate and select a reason (e.g., author, reviewer, approval) for critical actions, storing the signature in an immutable, audited ledger.

Business Value: Enables completely paperless approvals, saving days in lead candidate selection and workflow validations.

Scientific Value: Attaches a validated scientist signature to target validations, docking runs, and ADMET reports to confirm scientific peer review.

Navigation Instructions: Prompts automatically when signing off on workflows, registering molecules, or exporting compliance reports.

Screen Overview: A modal dialog overlay displaying the action details, signing reasons dropdown, password input field, and verification buttons.

Component Explanation:

Component / UI Element	Description & Functional Actions
Signer Profile Details	Displays the current user's name, email, and digital certificate details.
Signing Reason Dropdown	A mandatory selection of standard reasons: 'Author', 'Reviewer', 'Approval', or 'Sponsor'.
Electronic Signature Password Field	Enforces password re-entry to validate the signature identity.
Submit Signature Button	Applies cryptographic signature and appends it to the target entity.

Step-by-Step Workflow Steps:

1. Initiate a critical action (e.g., click 'Approve Workflow' in the designer).
2. Review the popup E-Signature modal.
3. Select 'Approval' as the signing reason.
4. Type your account password in the electronic password input field.
5. Click 'Sign and Submit'.
6. Verify the green 'Signature Applied' checkmark.

Expected Results: Successful execution of the signature, creation of an audit log containing the digital certificate details, and completion of the target action.

Errors & Troubleshooting:

Common Error Scenario	Troubleshooting Solution
Invalid Credentials	Password incorrect. Re-enter password. Note: 3 failures locks the digital signing capability.
Expired Certificate	User certificate is expired. Contact IT to renew your digital signing certificate.

Tips & Best Practices: Always check that your digital signature certificate is updated before entering critical IND validation phases.

Module 13: Data Lineage Explorer

Purpose: Visualizes the life cycle and movement of data across the platform, mapping out how raw assay values and molecular files were transformed, query-filtered, and analyzed into final candidates.

Business Value: Allows teams to trace data discrepancies back to source systems, resolving data quality issues and avoiding wasted lab work.

Scientific Value: Guarantees reproducibility by mapping data flows from ELN imports to query engines, regression curves, and target structures.

Navigation Instructions: Open the 'Compliance Console' from the left sidebar and select the 'Data Lineage' tab.

Screen Overview: An interactive flowchart rendering node connections representing data objects (files, tables, queries, models) and edge arrows representing data operations (import, filter, fit, export).

Component Explanation:

Component / UI Element	Description & Functional Actions
Lineage Canvas	Interactive ReactFlow board where users pan, zoom, and select lineage nodes.
Node Inspector Panel	Renders metadata for the selected node (e.g., date created, source system, execution parameters, owner).
Drill-Down Button	Opens the original service module that generated the data point (e.g., opens the specific query run).
Source Trace Button	Highlights the entire path of nodes back to the raw source data connector.

Step-by-Step Workflow Steps:

1. Open the Data Lineage Explorer.
2. Select the target node 'Lead Candidate Assay Fit' on the canvas.
3. Click 'Trace Source' in the inspector.
4. Observe the highlighted upstream path showing: Raw SQL Database Connector -> Query Builder AST -> 4PL Model Input -> Final Curve Output.
5. Double-click the 'Query Builder AST' node to inspect the original query rules.

Expected Results: A visual flow diagram illustrating data history, and displaying transformation details in the sidebar panel.

Errors & Troubleshooting:

Common Error Scenario	Troubleshooting Solution
Orphaned Node Reference	Occurs if an upstream raw data file was deleted from the server filesystem. The node appears yellow with a 'Miss
Lineage Indexing Latency	New data runs might take up to a minute to index in the lineage database. Refresh the lineage canvas.

Tips & Best Practices: Export the lineage flowchart as an image and include it in IND filings to demonstrate raw data traceability.

Module 14: AI Scientist Copilot

Purpose: An advanced natural language interface designed to interpret scientific questions, compile them to valid SQL or RDKit operations, query live databases, and return formatted tables, summaries, and structural charts.

Business Value: Accelerates data discovery and training times. Allows scientists to query complex informatics systems without writing queries or SQL.

Scientific Value: Leverages chemical and bioinformatics grounding rules to prevent model hallucinations, ensuring all answers represent actual database records.

Navigation Instructions: Click 'AI Scientist Copilot' or 'Copilot' in the left-hand sidebar menu, or go to '/copilot'.

Screen Overview: A chat-style panel containing recent conversations, an input text area for entering questions, and a right-hand workspace displaying tables and RDKit structures generated by the AI.

Component Explanation:

Component / UI Element	Description & Functional Actions
Chat Input Field	A text area where scientists enter natural language queries (e.g. 'What compounds target EGFR?').
Response Message Feed	Displays the conversation log, detailing the AI's explanation, generated SQL, and citations.
Visual Results Panel	Displays interactive data tables and molecular structure drawings resulting from the query.
Grounding Details Button	Expands to show the validation checks performed on the SQL before execution.

Step-by-Step Workflow Steps:

1. Open the AI Scientist Copilot page.
2. Type 'What compounds target EGFR?' in the input field.
3. Click the 'Send' button or press Enter.
4. Review the AI's explanation and inspect the generated SQL.
5. Observe the resulting table of active compounds in the results panel.
6. Ask a follow-up query: 'Filter this list for molecular weight less than 500'.

Expected Results: A natural language summary, a table of compounds matching the target query, structural drawings, and an audit trail reference.

Errors & Troubleshooting:

Common Error Scenario	Troubleshooting Solution
Hallucination Blocked	The AI generated SQL that failed structural verification. The query is automatically rejected, and a safe fallback query is provided.
Ambiguous Column Name	The query matches multiple columns in different tables. Clarify your question by specifying 'compound molecular weight' or 'assay molecular weight'.

Tips & Best Practices: Refer to the Prompts Library in the Copilot side panel for tips on structuring complex multi-table questions.

Module 15: Scientific Search

Purpose: A unified search portal that integrates keyword indexing, semantic search, and chemical structure searches into a single query tool.

Business Value: Reduces time spent looking for documents, protocols, or compound records across multiple platforms.
Scientific Value: Combines text annotations (e.g., target names, PubMed references) with structural searches (SMILES) and sequence patterns.

Navigation Instructions: Accessible from the header search bar or by clicking the 'Search' icon in the left-hand navigation sidebar.

Screen Overview: A clean search page centered around a search box, with filtering options for Category (All, Compounds, Sequences, Assays, Workflows) and Source (LIMS, DB, ELN).

Component Explanation:

Component / UI Element	Description & Functional Actions
Search Input Bar	Presents a search box supporting keywords, SMILES, sequence motifs, and BOOLEAN search logic.
Facet Filter Panel	Refines results by category, data source, date range, and owner.
Interactive Results List	Displays matching records with preview snippets, chemical structures, and links to detail pages.
Export Search Results Button	Exports search results to CSV or Excel formats.

Step-by-Step Workflow Steps:

1. Open Scientific Search.
2. Enter the keyword 'EGFR' and click Search.
3. Review the list of matching targets, compounds, and protocols.
4. Select the 'Compounds' facet from the side panel to filter results.
5. Double-click a compound card in the results to open its Compound Explorer page.

Expected Results: Renders matching items across different modules, with direct navigation links to details pages.

Errors & Troubleshooting:

Common Error Scenario	Troubleshooting Solution
Index Synchronization Lag	Newly added compounds may take up to 5 minutes to appear in keyword searches. Use direct search tools in the
Malformed Search Syntax	Incorrect Boolean syntax. Use quotes for exact phrases (e.g. 'EGFR assays') or separate terms with 'AND'.

Tips & Best Practices: Combine keywords with structure drawings to locate specific assay records for a particular chemical class.

Module 16: Project Management

Purpose: Organizes scientific studies, compounds, assays, and workflows into workspaces, facilitating collaboration within research teams.

Business Value: Protects sensitive research projects by restricting access to authorized project members.
Scientific Value: Maintains data organization across studies, linking targets to assays, compounds, and results to track progress.

Navigation Instructions: Accessible from the sidebar menu under the 'Projects' link.
Screen Overview: Displays active projects, recent updates, member lists, and study timelines.

Component Explanation:

Component / UI Element	Description & Functional Actions
Project Grid Panel	Cards showing project name, code (e.g., PROJ-EGFR-2026), status, and active member count.
Add Project Button	Launches a dialog to create a project, set privacy levels, and invite team members.
Project Workspace Tabs	Organizes project-specific resources: 'Compounds', 'Assays', 'Workflows', and 'Members'.
Timeline Chart	Visualizes milestone deadlines and study durations.

Step-by-Step Workflow Steps:

- 1. Navigate to the Projects page.
- 2. Click 'Create Project'.
- 3. Enter the project name 'KRAS Inhibitor Lead Group'.
- 4. Set the project visibility to 'Private - Invites Only'.
- 5. Add team members and assign roles (e.g., Scientist, Admin).
- 6. Click 'Save Project'.

Expected Results: Creation of the project space, updated navigation sidebar, and automated invitations sent to team members.

Errors & Troubleshooting:

Common Error Scenario	Troubleshooting Solution
Duplicate Project Code	Project code already exists. Enter a unique identifier for the study.
Insufficient User Rights	Only project administrators can modify project membership or settings.

Tips & Best Practices: Link all related queries and workflow runs to a project workspace to keep study data consolidated.

Module 17: User Administration

Purpose: Provides user management capabilities, allowing administrators to invite users, activate/deactivate accounts, monitor active sessions, and assign system access levels.

Business Value: Maintains system security by ensuring only authorized personnel can access sensitive discovery data.
Scientific Value: Enforces regulatory compliance by ensuring user details are linked to electronic signatures and audit trails.

Navigation Instructions: Go to the Settings panel and select the 'User Admin' tab (requires Administrator role).

Screen Overview: A dashboard displaying the user directory, account status badges, and action buttons.

Component Explanation:

Component / UI Element	Description & Functional Actions
User Directory Table	Lists user name, email, role, status (Active, Suspended, Invited), and last login time.
Invite User Button	Launches a modal to send email invitations and pre-assign roles.
Account Status Toggle	Enables or disables accounts instantly.
Session Activity Monitor	Displays active browser sessions with IP addresses and logout actions.

Step-by-Step Workflow Steps:

- 1. Open Settings and select the 'User Admin' tab.
- 2. Click 'Invite User'.
- 3. Enter the user's email: 'm.curie@analytix.com'.
- 4. Select the 'Scientist' role from the dropdown.
- 5. Click 'Send Invitation'.

Expected Results: A notification confirming the invitation has been sent, and an entry added to the directory table with an 'Invited' status.

Errors & Troubleshooting:

Common Error Scenario	Troubleshooting Solution
User Already Registered	Email address is already registered. Edit the existing user's profile to change their settings.
Domain Verification Failure	Email domain is not on the whitelist. Verify email address or contact security team.

Tips & Best Practices: Review the user directory monthly to deactivate inactive accounts and maintain license compliance.

Module 18: Role Based Access Control (RBAC)

Purpose: Defines access control matrices that link roles (Admin, Scientist, Compliance Officer, Read-Only) to specific system permissions, ensuring users only access authorized modules.

Business Value: Prevents unauthorized data modification and protects IP, ensuring compliance with data security requirements.

Scientific Value: Prevents accidental modification of shared templates, assay records, and compliance logs by restricting access to authorized users.

Navigation Instructions: Go to the Settings panel and select the 'RBAC Control' tab (requires Administrator role).

Screen Overview: An interactive matrix mapping roles to system permissions, with custom role creation tools.

Component Explanation:

Component / UI Element	Description & Functional Actions
RBAC Matrix Grid	A grid where columns represent roles and rows represent system actions, with checkboxes to manage permissions.
Custom Role Creator	Launches a panel to define a role name and select its permissions.
Policy Editor Panel	Sets conditional access rules, such as login IP whitelists and access windows.
Save RBAC Map Button	Saves the access control matrix and updates system permissions.

Step-by-Step Workflow Steps:

1. Open Settings and select the 'RBAC Control' tab.
2. Locate the 'Scientist' role column in the matrix.
3. Check the box for 'Metadata Schema Edit' to grant editing permissions.
4. Uncheck the box for 'Audit Log Delete' to restrict deletion rights.
5. Click 'Save RBAC Map' and enter your electronic signature password.

Expected Results: An audit log entry documenting the permission change, and updated access rights applied to all affected accounts.

Errors & Troubleshooting:

Common Error Scenario	Troubleshooting Solution
Permission Conflict	Invalid combination of permissions (e.g. read-only role with editing rights). Verify changes in the validator panel.
Cannot Demote Self	Administrators cannot revoke their own admin permissions. Have another admin perform the action.

Tips & Best Practices: Follow the principle of least privilege. Grant write permissions only when necessary for a user's role.

Module 19: Notifications & Alerts

Purpose: A notification center that alerts users to workflow updates, data sync status, connection changes, and compliance reviews.

Business Value: Keeps project teams aligned on study progress, reducing delays in review and sign-off stages.
Scientific Value: Alerts researchers when docking simulations, ADMET calculations, or sequence alignments are complete.

Navigation Instructions: Accessible by clicking the 'Bell' icon in the top header, or going to the Settings page and selecting 'Alerts Config'.

Screen Overview: A list of recent notifications with category filters and configuration panels for email and Slack integrations.

Component Explanation:

Component / UI Element	Description & Functional Actions
Notifications List	A panel showing messages, source modules, and action links.
Integration Panel	Configures webhook settings for Slack, Microsoft Teams, and email notifications.
Trigger Rules Table	Lists notification rules, allowing users to select which events trigger alerts.
Mark All Read Button	Clears all active notifications instantly.

Step-by-Step Workflow Steps:

1. Click the 'Bell' icon in the header.
2. Select a notification showing 'Workflow Execution Complete' to view results.
3. Go to settings to configure notifications.
4. Toggle 'Email Alerts' to enabled for 'Compliance Verification Failures'.
5. Click 'Save Notification Preferences'.

Expected Results: Updated notification settings and confirmation that alerts will be delivered via the selected channels.

Errors & Troubleshooting:

Common Error Scenario	Troubleshooting Solution
Delivery Failure	The email server or webhook returned an error. Verify notification settings in the Settings tab.
Notification Blocked	Browser notifications are blocked by user settings. Enable notifications in your browser settings.

Tips & Best Practices: Configure email digests for routine events to minimize inbox noise, while keeping real-time alerts enabled for critical compliance failures.

Module 20: Settings

Purpose: The main settings panel containing configuration settings for user profiles, theme options, security preferences, database credentials, and network configurations.

Business Value: Enables administrators to configure the platform to meet organizational security and network requirements.

Scientific Value: Allows researchers to customize molecular drawing preferences, sequence alignment parameters, and search defaults.

Navigation Instructions: Select 'Settings' from the left sidebar or navigate to '/settings' (accessible via User Profile footer).

Screen Overview: A multi-section settings dashboard organized into: User Settings, Security & Auth, Connectors, and System Config.

Component Explanation:

Component / UI Element	Description & Functional Actions
Password Policy Panel	Sets password requirements, including minimum length, character types, and expiration rules.
SMTP Settings Form	Configures email server details, including host, port, username, password, and encryption protocol.
Proxy Configuration Fields	Sets proxy details for systems that require proxy access to external web services.
Database Maintenance Panel	Allows administrators to run database optimization tasks, prune logs, and back up schemas.

Step-by-Step Workflow Steps:

1. Open Settings from the sidebar.
2. Go to the 'Security & Auth' tab.
3. Change password complexity requirement to 'Minimum 12 Characters'.
4. Go to 'System Config' and enter SMTP server settings: SMTP Host 'smtp.analytix.com', Port '587'.
5. Click 'Save Configurations' and enter your electronic signature password.

Expected Results: Updated system settings and a confirmation message indicating configurations have been applied successfully.

Errors & Troubleshooting:

Common Error Scenario	Troubleshooting Solution
Invalid SMTP Credentials	Connection to SMTP server failed. Verify hostname, port, and credentials.
Save Failed	Missing mandatory configuration values. Complete all required fields before saving.

Tips & Best Practices: Back up system configurations before making changes to settings or database connections.

5. AI Scientist Copilot Prompt Guide

The AI Scientist Copilot translates natural language questions into database operations using grounding rules. This section lists prompt examples and how they are parsed by the system:

Prompt: *"What compounds target EGFR?"*

System Parsing & Grounding: Runs a join query between compounds and assay targets, returning a list of chemical structures and their corresponding IC50 values against the epidermal growth factor receptor.

Prompt: *"Show compounds with MW < 500 and LogP < 5"*

System Parsing & Grounding: Performs a visual query filtering the database for drug-like small molecules matching key criteria of Lipinski's Rule of 5.

Prompt: *"Summarize assay results"*

System Parsing & Grounding: Aggregates screening data to report standard metrics (e.g., active hits, average IC50, total compounds tested) across recent assay runs.

Prompt: *"Find compounds active against KRAS"*

System Parsing & Grounding: Queries the database for compounds exhibiting active status or low nanomolar potency against KRAS targets.

Prompt: *"Show workflow execution history"*

System Parsing & Grounding: Retrieves log entries from the workflow schema, listing the history of scientific pipeline runs, durations, and execution statuses.

6. Scientific Analytics Workbench Documentation

The Analytics Workbench provides advanced statistical modeling tools. It processes query results and performs calculations, including 4-Parameter Logistic (4PL) regression, Principal Component Analysis (PCA), and molecular similarity matrices.

6.1 Analytics Metrics Reference Table

Analytics Category	Core Metrics	Scientific Calculations	Visualization Outputs
Dashboard KPIs	System throughput, compound registry	Aggregate counts, daily rates	Time-series charts
Compound Analytics	MW, LogP, Tanimoto similarity	Molecular descriptors, fingerprint similarity	Scatter plots, similarity matrices
Assay Analytics	EC50, IC50, Hill slope, R2	4PL curve fitting, non-linear regression	Sigmoidal response curves
Bioinformatics	Alignment scores, sequence identity	Smith-Waterman score, clustering distance	Heatmaps, dendrograms
Workflow	Execution duration, node success	Resource completion, average runtime	Flow progress timelines
Audit & Compliance	Log counts, ledger status	Verification calculations, hash check	Audit charts, integrity bars
AI Usage	Query volume, validation rate	Accuracy percentage, prompt count	Usage charts, token volumes

7. Compliance & FDA Validation

Operating in a regulated pharmaceutical environment requires compliance with FDA 21 CFR Part 11 regulations. AnalytiX enforces these rules across the system.

Audit Logs Chaining: All logs are secured using SHA-256 hash chains. Modifying a log entry invalidates the chain, alerting system administrators during ledger checks.

Electronic Signatures Double-Factor: Critical actions (e.g., compound approval, data export) require users to re-enter their passwords and specify a reason for signing.

Version Control: Workflows, query templates, and metadata structures are versioned. Previous versions are archived to maintain audit trails.

Data Lineage Traceability: The system logs transformation steps, mapping files and query metrics back to source connectors for full traceability.

8. Troubleshooting FAQs

Q: How do I resolve 'RDKit Parse Error' when registering a compound?

A: Ensure the SMILES string you entered has valid chemical valency and correct formatting. Use the JSME sketcher to draw the structure and automatically generate the SMILES string.

Q: Why can't I see the 'User Admin' or 'RBAC Control' tabs in Settings?

A: These tabs are restricted to users with the Administrator role. Contact your system administrator to request access if required.

Q: What should I do if the 'Ledger Integrity' check reports a warning?

A: Contact your Compliance Officer or security team immediately. A warning indicates that audit logs have been altered outside the platform interface.

Q: How do I download a workflow run execution report?

A: When a workflow run completes, click the 'Export Report' button in the execution drawer to download a PDF report containing the run logs and results.

Q: Why did my molecular docking workflow run time out?

A: Molecular docking tasks can be computationally intensive. Open the node configuration drawer and reduce the docking grid size or adjust execution parameters.

9. Platform Glossary

SMILES: Simplified Molecular-Input Line-Entry System. A line notation for describing the structure of chemical species using short ASCII strings.

FASTA: A text-based format for representing either nucleotide sequences or peptide sequences, in which nucleotides or amino acids are represented using single-letter codes.

RDKit: An open-source cheminformatics and machine learning software toolkit written in C++ and Python.

4PL Regression: 4-Parameter Logistic regression. A mathematical model used to analyze sigmoidal dose-response curves in biological assays.

FDA 21 CFR Part 11: Title 21 of the Code of Federal Regulations; Electronic Records; Electronic Signatures. Enforces requirements for closed and open systems to ensure record authenticity.

Tanimoto Coefficient: A statistical metric used to quantify the similarity between two molecular fingerprints, ranging from 0.0 (completely different) to 1.0 (identical).

EAV Model: Entity-Attribute-Value model. A database design pattern used to represent entity properties dynamically without modifying the database schema.

AST Graph: Abstract Syntax Tree. A tree representation of the abstract syntactic structure of source code or visual queries.

SHA-256: Secure Hash Algorithm 2. A cryptographic hash function that generates a unique 256-bit signature for data block verification.

IND: Investational New Drug. An application filed with the FDA to obtain authorization for clinical testing of a new drug candidate.